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2027 CYCLE

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CUET UG

Biology

Ecosystem

Official Syllabus Unit X · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit X)	Skills the questions test
Structure and function	Structure and function.	Naming the parts of an ecosystem and the job each part does.
Productivity	Productivity.	Computing NPP from GPP and respiration; telling a rate from a stored amount, with units.
Decomposition	Decomposition.	Listing the 5 steps in order, humus properties, and what speeds or slows decay.
Energy flow — trophic levels	Energy flow – different trophic levels.	Placing organisms at levels and applying the 10% transfer rule up the ladder.
Energy flow — food chain and food web	Energy flow – food chain and food web.	Telling a grazing chain from a detritus chain, reading webs, explaining why chains stay short.
Ecological pyramids — of energy, biomass and number	Ecological pyramids – pyramid of energy, biomass and number.	Judging which pyramids stay upright, which can invert, and why.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit X poses ecology's working questions: how energy enters an ecosystem, where it goes, and why it shrinks on the way.

What an Ecosystem Is, and How to Read One

Drop a few waterweeds into a jar of pond water, add a snail, and leave the jar in the sun. The plants make food, the snail eats them, and bacteria quietly tidy up everything that dies. Nothing more needs to go in except sunlight. That jar is a tiny working model of ecology's biggest idea:

the **ecosystem** — a unit of nature where living things and their physical surroundings interact as one system.

- **The definition:** an ecosystem is all the organisms in an area together with their physical environment, interacting as one working system — a pond, a forest, a grassland, even a rotting log.
- **Two halves:** the **abiotic** part is the non-living setting — water, air, soil, sunlight, minerals. The **biotic** part is every living thing inside it.
- **Interaction is the point:** organisms soak up water and nutrients, feed on one another, and return matter to the soil. Remove any one of those moves and the system stops working.
- **Function, not just structure:** like any machine, an ecosystem does four jobs — it captures energy, makes food, passes food along, and recycles dead matter.

The pond: NCERT's own worked example

- **Abiotic inputs:** water, dissolved minerals, sunlight from above, and a soft sediment of mud at the bottom.
- **Producers:** green **phytoplankton** — microscopic floating algae that photosynthesise — plus rooted water plants.
- **Herbivores: zooplankton** (tiny drifting animals), snails and small fish nibble the producers.
- **Carnivores:** bigger fish hunt the small fish; dragonfly young hunt whatever swims too close.
- **Decomposers:** bacteria and fungi break every dead body and sunken leaf back into dissolved nutrients.
- **The loop closes:** those released nutrients dissolve back into the water, where the phytoplankton pick them up again. The pond feeds itself.

Reading the pond in layers: stratification

- **Species composition** means exactly which species live in the ecosystem, and how many of each — the pond's cast list.
- **Stratification** means the vertical **layering** of that life, one floor above another.
- **In a forest:** tall trees form the canopy, shrubs sit under them, herbs carpet the floor.
- **In a pond:** floating phytoplankton ride the sunlit surface, submerged weeds sway deeper down, and worms and microbes work the bottom mud.
- **Why layers matter:** each layer is a separate address. Two species can share one pond peacefully because they live at different depths.

Role	Job in the system	Pond example
Producers	Trap sunlight and build organic matter — living plant tissue.	Phytoplankton, water weeds
Herbivores (primary consumers)	Eat living producers directly.	Zooplankton, snails

Role	Job in the system	Pond example
Carnivores (secondary consumers and above)	Eat other animals, one rung higher each time.	Large fish, dragonfly young
Decomposers	Break dead matter into simple, reusable nutrients.	Bacteria, fungi

Why the pond is the perfect "mini ecosystem"

- **Small enough to hold:** you can watch every step, from sunlight to snail to sediment, in a single jar.
- **All four roles on show:** producers, herbivores, carnivores and decomposers all live within arm's reach.
- **Self-sustaining:** only sunlight crosses the boundary from outside; everything else circulates inside.
- **A scale model:** a forest or ocean runs on the same four roles with a longer cast — learn the pond, and you can read any ecosystem.

EVERYDAY EXAMPLE — THE COLLEGE CANTEEN

A canteen kitchen works like an ecosystem with four posts. The **cooks** are the producers: they convert electricity (sunlight) and raw stock into meals. The **students** are the consumers, eating one another's leftovers in a queue — seniors first. The **cleaning crew** are the decomposers, scraping plates back into the store room as reusable stock. Notice the honest part of the map: **ingredients cycle** — yesterday's scraps restock tomorrow's meals — but the **electricity flows once** and is gone. Ecosystems work the same way.

Productivity — the Ecosystem's Income

A family earns a salary every month and burns through part of it just to keep the household running — travel, electricity, groceries. What survives those running costs is what they can actually spend on anything new. Ecosystems keep accounts in exactly the same shape: producers earn energy income by photosynthesis, pay a running cost called respiration, and the balance is the only food every animal in the chain will ever get.

- **Gross primary productivity (GPP):** the **rate** at which producers capture sunlight and build new organic matter — plant tissue — per unit area per year.
- **Respiration (R):** the producers' running cost — they burn part of the food they just made to stay alive, releasing that energy as heat.
- **Net primary productivity (NPP):** what is left after the running cost — the only energy available to every consumer and decomposer in the ecosystem.
- **Secondary productivity:** the rate at which consumers build their own new organic matter out of the food they eat.

THE PRODUCTIVITY RELATION

$$\text{NPP} = \text{GPP} - R$$

- R = respiration — the food producers burn to stay alive
- "NPP" is the share that actually reaches the animals

Rate, not stock: units and the two "standing" terms

- **Units:** GPP and NPP are measured in g/m²/yr or kJ/m²/yr. Read the "/yr" — productivity is a **rate**, like a monthly salary.
- **Standing crop:** the mass of living material present at one moment — that is a **stock**, like a bank balance.
- **Standing state:** the amount of inorganic nutrients sitting in the soil at one moment — also a stock, but of dead, non-living nutrients.
- **The mix-up:** productivity flows; standing crop sits. One question can put both words in a single line, so keep them apart.

Point	Standing crop	Standing state
What it measures	Mass of living material now	Amount of inorganic nutrients in soil now
Living or dead	Living — organisms, alive or recent	Dead — simple inorganic forms
Sits where	In the bodies of organisms	In the soil and water
Example	2 kg of fish per m² of pond	Nitrate level of the pond mud
Rate or stock	A stock — one frozen moment	A stock — one frozen moment

Where the oceans stand

- **Big area, modest output:** oceans cover about **70% of Earth's surface**, yet contribute a **smaller share** of the world's total productivity.
- **Why the sea underperforms:** light dies out with depth, and dissolved nutrients run scarce — so per unit area, ocean water is a poor farm.
- **The contrast to remember:** a hectare of rich farmland out-produces a hectare of open ocean many times over.

WORKED EXAMPLE — FINDING NPP

A grassland's producers build new matter worth GPP = 20,000 kJ/m²/yr and respire R = 8,000 kJ/m²/yr. Find NPP and the share available to consumers.

$$\text{NPP} = \text{GPP} - R$$

$$\text{NPP} = 20,000 - 8,000$$

$$\text{NPP} = 12,000$$

So NPP = **12,000 kJ/m²/yr** — this is the only part the herbivores can ever eat.

- **Share reaching consumers:** 12,000 ÷ 20,000 = 0.6, so **60%** of GPP survives.
- **Check the logic:** if respiration were 0, NPP would equal GPP — respiration is the only subtraction in the relation.

Quick drill: NPP from GPP and R

GPP (kJ/m²/yr)	R (kJ/m²/yr)	NPP (kJ/m²/yr)
20,000	8,000	12,000
35,000	15,000	20,000
9,000	6,000	3,000

- **How to drill it:** cover the third column, subtract aloud — GPP minus R each time.
- **Watch the units:** every column carries kJ/m²/yr, because all three are rates.
- **Exam instinct:** if a question hands you GPP and R, the NPP line is one subtraction away — the rest of the question builds on that number.

Decomposition — Nature's Recycling Line

A healthy forest drops several tonnes of leaves every year. Yet walk through it and the floor is a thin carpet, not a mountain of dead matter. Something removes the dead material as fast as it arrives and hands it back to the soil as nutrients. That cleanup-and-recycling operation is **decomposition**, and it runs on five fixed steps in a fixed order.

- **Detritus:** the raw material — dead plant remains such as fallen leaves, plus dead animal bodies and droppings.
- **Detritivores:** the animals that feed on detritus — the earthworm is the classic example.
- **Decomposition:** the whole process by which complex dead matter is broken down into simple inorganic substances the soil can reuse.
- **Oxygen is fuel:** the line runs on oxygen; shut off the air and it slows to a crawl.

The five steps, in order

1. **Fragmentation:** detritivores chew large pieces of detritus into smaller bits, exposing more surface for the microbes.
2. **Leaching:** water dissolves the soluble substances and carries them down into the soil.
3. **Catabolism:** bacteria and fungi pour out enzymes that chemically break the small bits into simpler compounds.
4. **Humification:** what remains settles into **humus** — a dark, amorphous (shapeless) substance.
5. **Mineralisation:** further microbial action degrades the humus and releases inorganic nutrients back to the soil.

Step	What happens	Who or what does the work
1. Fragmentation	Large detritus broken into small pieces.	Detritivores, e.g. earthworms
2. Leaching	Soluble substances washed downward into the soil.	Rainwater seeping through
3. Catabolism	Complex matter chemically broken into simpler compounds.	Bacterial and fungal enzymes
4. Humification	Leftover matter turns into dark, amorphous humus.	Microbial action
5. Mineralisation	Humus degrades further, releasing inorganic nutrients.	Soil microbes

Humus: the soil's fixed deposit

- **Dark and amorphous:** humus has no fixed shape or structure of its own.
- **Slow to decay:** it resists microbial attack, so it can sit in the soil for years.
- **A nutrient reservoir:** it holds nutrients and releases them slowly — the soil's long-term savings account.

What speeds it up, what slows it down

- **Warm and moist = fast:** tropical forest litter can vanish within weeks.
- **Waterlogged = slow:** anaerobic conditions (no oxygen) drag the line — leaves in waterlogged bogs stay readable for centuries.
- **Chemical quality of detritus:** sugar-rich material disappears quickly; **lignin** (wood fibre) and **chitin** (insect shells) take years.
- **Oxygen supply:** loose, aerated soil feeds the microbes; sealed, soggy layers stall them.

Productivity vs decomposition — do not mix them up

One builds matter up, the other breaks matter down. A question can place the two words in one line, so pin the contrast:

Point	Productivity	Decomposition
What happens	New organic matter is built from light.	Dead matter is broken into simple nutrients.
Direction	Builds up — storage.	Breaks down — release.
Energy role	Stores sunlight as chemical energy.	Leftover energy finally leaves as heat.
Who runs it	Green producers.	Detritivores, bacteria, fungi.

Point	Productivity	Decomposition
What to quote	A rate — g/m²/yr or kJ/m²/yr.	The 5 steps, in their fixed order.

Order them — a 60-second drill

Say the five steps aloud before checking the answer line below. Shuffled for practice: catabolism, fragmentation, mineralisation, leaching, humification.

- **Correct order:** fragmentation → leaching → catabolism → humification → mineralisation.
- **Why this order works:** chew first (fragmentation), wash second (leaching), digest third (catabolism), settle fourth (humification), release last (mineralisation).
- **Exam instinct:** if a question swaps leaching and catabolism, reject it — water carries the dissolved bits down before the enzymes start their chemical work.

EVERYDAY EXAMPLE — THE COMPOST BIN ON THE BALCONY

A compost bin is decomposition in a box, step by step. The dry leaves and peels you drop in are the **detritus**; the worms that arrive are **detritivores**, doing fragmentation. The brown liquid dripping out of the bottom is exactly what **leaching** means — soluble stuff carried down by water. The heat inside a healthy pile is **catabolism** at work. The dark, crumbly finished compost is **humus-like**, and feeding it to your pots is **mineralisation** — nutrients back to the roots. Keep the pile damp, warm and turned for speed; seal it in a plastic bag and it just stinks — that is the anaerobic slowdown.

Energy Flow — the One-Way Street

Watch a waterfall: the water goes over the edge once and never climbs back. Energy in an ecosystem behaves the same way — it arrives as sunlight, climbs the food chain step by step, and leaves as heat. **Nutrients recycle; energy never does.** Hold on to that one contrast and the rest of this chapter falls into place.

- **The Sun is the only source:** nearly every unit of energy in every food chain began as sunlight.
- **PAR:** photosynthetically active radiation — the wavelengths of light that chlorophyll can actually absorb. Less than half of incident sunlight, about 50%, is PAR.
- **Trophic level:** one step of a food chain, numbered by how far the energy has travelled. Producers sit at level 1 (T1), herbivores at T2, carnivores at T3.
- **Only about 10% climbs:** of the energy entering any level, roughly a tenth ends up stored in the bodies of the next level.

THE 10 PERCENT RULE (LINDEMAN)

$$E_{n+1} \approx \frac{10}{100} \times E_n$$

- E_n = energy entering trophic level n (producers are level 1)
- E_{n+1} = energy that reaches the next level up
- The other 90% leaves the chain — spent as heat

Two chains, one ecosystem

- **Grazing food chain (GFC):** starts from **living producers** — grass → rabbit → fox.
- **Detritus food chain (DFC):** starts from **dead matter** — fallen leaves → detritivores and decomposers → the predators that eat them.
- **The chains connect:** DFC animals become prey for GFC hunters, so energy leaks across from one chain into the other.
- **Forests lean on the DFC:** most forest production ends as litter on the floor, so the detritus route carries the bigger share of the energy there.

Point	Grazing chain (GFC)	Detritus chain (DFC)
Starts with	Living green producers	Dead organic matter (detritus)
First eaters	Herbivores	Detritivores and decomposers
Example	Grass → rabbit → fox	Leaf litter → earthworm → bird
Bigger share where	Open grasslands, ponds	Forests, where litter piles up

Why a food web, not a chain

- **One-to-many reality:** a rabbit feeds a fox, a hawk and a human — no organism has exactly one predator and one prey.
- **Food web:** many chains interlocked into one map of who eats whom. Real ecosystems are webs; the single straight chain is a classroom sketch.
- **A stability bonus:** if one food source fails, a web-eater switches to another, and the ecosystem keeps running.

Energy moves, nutrients cycle

Question	Energy	Nutrients
Direction	One-way — up the chain, then lost	Cycle endlessly between soil and life
At each transfer	About 90% leaks out as heat	Returned to soil by decomposers
Does it return?	Never — no path back to producers	Always — the loop is closed
Refuelled by	Fresh sunlight every day	Nothing needed — recycling is self-driven

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WORKED EXAMPLE — CLIMBING THE 10% LADDER

Suppose the producers of a grassland patch store 10,000 kJ/m²/yr of new plant matter (they captured about 1,00,000 kJ of sunlight to build it). Follow the energy upward, keeping only 10% at each step:

$$10,000 \times \frac{10}{100} = 1,000$$

$$1,000 \times \frac{10}{100} = 100$$

$$100 \times \frac{10}{100} = 10$$

Level	Who sits here	Energy (kJ/m²/yr)
T1	Grass plants (producers)	10,000
T2	Herbivores	1,000
T3	Carnivores	100
T4	Top carnivores	10

- **Check:** every row is exactly 10% of the row below it.
- **Why chains stay short:** a fifth level would hold only 1 kJ/m² — far too little to keep any population alive. Real chains stop at 3–4 levels; energy, not appetite, sets the limit.

Ecological Pyramids — Three Lenses

Slice any ecosystem by trophic level and stack the slices, producers at the bottom. What you place inside each slice — a headcount, a dry weight, or stored energy — gives the stack its name and its shape. Ecologists call these stacks **ecological pyramids**, and each kind has its own rules about when it stands upright and when it turns upside down.

- **The base rule:** producers always form the bottom bar; each bar's width shows how much of the measured thing that level holds.
- **Pyramid of number:** counts the organisms at each level. A grassland gives countless grass plants, fewer herbivores, still fewer carnivores — upright.
- **The tree surprise:** one single tree (T1) can support hundreds of fruit-eating insects and birds (T2), so the pyramid of number turns **inverted** in a tree ecosystem.
- **Pyramid of biomass:** weighs the dry living mass at each level. On land this stays upright — plants outweigh their eaters.
- **The ocean surprise:** the standing crop of phytoplankton is small at any instant, yet it multiplies fast enough to feed a larger standing crop of zooplankton — an **inverted** biomass pyramid in the sea.
- **Pyramid of energy:** measures the energy stored in new tissue at each level. It is **always upright**, without exception.

- **Why energy never inverts:** the second law of thermodynamics — every transfer leaks energy as heat, so no level above can ever hold more energy than the level below.
- **Known blind spots:** a pyramid treats one species as belonging to one level only, and counts one big organism (the tree) as a single unit.

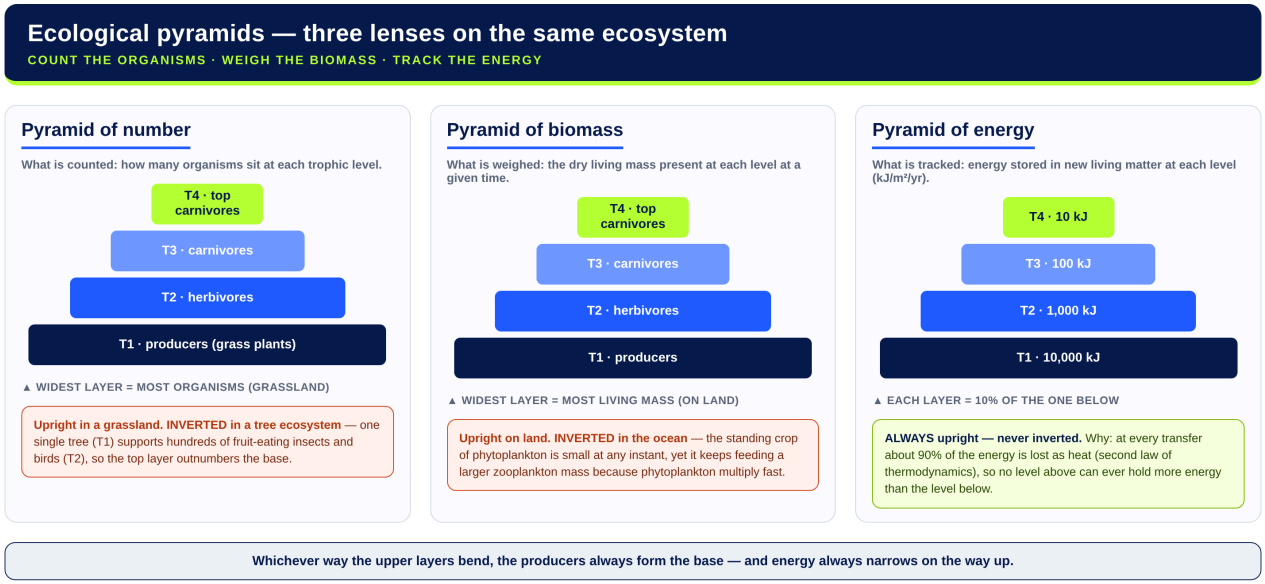


Figure 1: The three pyramids side by side — number can invert on a tree, biomass can invert at sea, energy never inverts.

Pyramid	What the bars hold	Can it invert?	Where — and why
Of number	Count of organisms per level	Yes	Tree ecosystem — one tree feeds hundreds of insects
Of biomass	Dry living mass per level	Yes	Ocean — tiny standing phytoplankton, larger zooplankton
Of energy	Energy stored in new tissue	Never	Heat escapes at every transfer (second law)

Two more shapes worth knowing

- **Parasitic food chain:** one large host (T1) carries many parasites (T2) — the pyramid of number inverts here for the same reason as the tree.
- **The single-plant rule:** whenever the base is one very large organism — a banyan tree, a mango tree — expect the count pyramid to flip.
- **A bulge, not a break:** if the middle level outnumbers both neighbours, the stack bulges in the middle; only the energy pyramid refuses every such shape.
- **Instant check:** name the ecosystem first, then pick the pyramid — the ecosystem decides the shape, never the other way round.

Quick drill: which shape is it?

Ecosystem	Pyramid of number	Pyramid of biomass
Grassland	Upright	Upright
Tree with its insects	Inverted	Upright
Ocean	Upright	Inverted
Parasitic chain	Inverted	Upright

- **Read the table twice:** the pyramid of biomass only flips where the **standing crop** of the base is tiny but fast-growing — the open ocean.
- **The energy column is missing on purpose:** it would say "upright" in every row.

Putting It Together

Watch one loop of the machine: light falls, plants earn, animals eat, decomposers clean up, the soil restocks, and the Sun must pay again tomorrow. The six lines below are the whole chapter in one breath — revise from them the night before the exam.

- **Sun → PAR:** less than half of incoming sunlight is light plants can use (PAR).
- **PAR → GPP:** producers turn that light into new organic matter at the GPP rate.
- **GPP → NPP:** subtract respiration; NPP is the food every consumer lives on.
- **NPP → consumers:** herbivores first, then carnivores, each keeping about 10% of what they receive.
- **All deaths → detritus:** the 5-step decomposition line takes over every death.
- **Nutrients → soil → producers:** matter cycles forever; energy exits as heat and must be re-earned from the Sun each day.

How to think about an ecosystem MCQ

1. **Is the pyramid inverted?** Number (tree ecosystem) and biomass (ocean) can invert; energy never does.
2. **Which step of decomposition?** Fix the order — fragmentation, leaching, catabolism, humification, mineralisation.
3. **Which way does energy move?** One-way and upward, leaving as heat — never back toward the producers.
4. **Rate or stock?** Productivity is per year; a standing crop is one frozen moment.
5. **Did you subtract respiration?** NPP = GPP − R — respiration is the only subtraction.

The chapter in numbers

Ten numbers carry the whole unit. If you can explain each line in one sentence, you are ready for the paper.

Number	What it tells you
Less than 50%	Share of incident sunlight that is PAR — the only part plants can use.
10%	Energy handed up from one trophic level to the next (Lindeman rule).
90%	Energy lost as heat at every transfer — the reason chains stay short.
20,000 − 8,000 = 12,000	The NPP subtraction: GPP minus respiration, in kJ/m²/yr.
5	Steps of decomposition — fragmentation, leaching, catabolism, humification, mineralisation.
3–4	Trophic levels a real food chain can hold before energy runs out.
About 70%	Earth's surface that is ocean — yet the sea contributes a smaller share of productivity.
g/m²/yr, kJ/m²/yr	Productivity units — always a rate, never a stock.
2 kg fish per m²	A standing crop — a stock measured at one frozen moment.
Never	The chance that a pyramid of energy inverts.

A day in the pond — every idea in one story

- **Morning:** sunlight strikes the surface; only the PAR half of it matters to the phytoplankton.
- **By afternoon:** the phytoplankton have built new matter at the GPP rate — and burned part of it as respiration to stay alive.
- **The balance:** GPP − R = NPP, the pond's real income for the day.
- **Evening meal:** zooplankton graze the producers and keep about 10% of what they eat; the fish that eat them keep 10% of that.
- **After dark:** a waterweed dies, sinks, and becomes detritus; the 5-step line takes over — fragmentation, leaching, catabolism, humification, mineralisation.
- **Back to the start:** released nutrients dissolve into the water and wait for tomorrow's sunlight.
- **The pond's pyramids:** number and biomass stand upright here; the energy pyramid stands upright everywhere, every single time.

True or false — five quick judges

- **"NPP is GPP minus decomposition."** False — subtract respiration only.
- **"The ocean can show an inverted pyramid of biomass."** True — tiny standing phytoplankton, larger zooplankton.
- **"A pyramid of energy can invert if the top predator is huge."** False — heat leaves at every transfer, so energy always narrows upward.

- **"Leaching is optional in the decomposition sequence."** False — it is step 2 of 5, right after fragmentation.
- **"Productivity of 500 g/m² is a complete answer."** False — a rate needs its "/yr" to be an answer at all.

Two-minute teacher dialogue

You ask	The teacher answers
Why is only 10% passed on?	Bodies burn most of what they eat for movement and warmth; only about a tenth lands in new tissue.
Where does the lost 90% go?	Out of the food chain as heat — the second law of thermodynamics at work.
Can nutrients run out like energy?	No — decomposers return every nutrient; only energy needs daily refuelling from the Sun.
Which pyramid should I trust in an MCQ?	Energy — upright in every ecosystem; the other two depend on where you look.
Why do ponds make good models?	All four roles live within arm's reach, and only sunlight crosses the boundary.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — ECOSYSTEM

Term / fact	What it means	Example / number
Ecosystem	Organisms + physical environment interacting as one system.	A pond; a forest
NPP relation	$\text{NPP} = \text{GPP} - R$ — one subtraction, respiration only.	$20{,}000 - 8{,}000 = 12{,}000$ kJ/m²/yr
Productivity units	A rate per unit area per year — never a stock.	g/m²/yr or kJ/m²/yr
PAR	The share of sunlight plants can actually use.	Less than half, about 50%
10% rule	Energy handed up to each next trophic level.	10,000 → 1,000 → 100 → 10 kJ
5 decomposition steps	The order never changes.	fragmentation → leaching → catabolism → humification → mineralisation

Term / fact	What it means	Example / number
Humus	Dark, amorphous, decays very slowly.	A reservoir of nutrients in the soil
Pyramid of number	Count per level; inverts for trees.	One tree → hundreds of insects
Pyramid of biomass	Dry mass per level; inverts at sea.	Phytoplankton → zooplankton
Pyramid of energy	Always upright — no exceptions.	Heat lost at every transfer
Standing crop vs standing state	Living mass now vs soil nutrients now.	2 kg fish per m^2 vs soil nitrate level

Two memory anchors: NPP is the only food animals will ever get, and the energy pyramid is the only pyramid that can never turn upside down.

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	Why / the trap
Writing NPP = GPP − decomposition	NPP = GPP − respiration (R).	Decomposition is a separate process; only respiration is subtracted.
Quoting productivity without a time unit	Always g/m^2/yr or kJ/m^2/yr — a rate.	No "/yr" means a stock, and the answer is wrong.
Treating the energy pyramid as invertible	The pyramid of energy is always upright.	The second law sends heat out at every step.
Listing only 3 decomposition steps	All 5: fragmentation, leaching, catabolism, humification, mineralisation.	Leaching sits quietly at step 2 and is easy to forget.
Calling the DFC the "clean, waste-free" chain	The DFC **is** the waste route — it starts from detritus.	In nature nothing is waste; detritus is someone's lunch.

The mistake students make	What is right	Why / the trap
Drawing one neat chain for a real ecosystem	Draw a food web — chains interlock.	One organism feeds many others and eats many others.
Using 10% as an exact, universal figure	10% is an approximate transfer efficiency.	Use it to compare levels, not as a guaranteed constant.

- **Subtract once:** NPP = GPP − R — respiration is the only subtraction in the chapter.
- **Say the units:** every productivity number needs its "/yr" to stay a rate.
- **Test the pyramid:** if the question says **energy**, the shape is upright — no exceptions.

The Whole Chapter on One Page

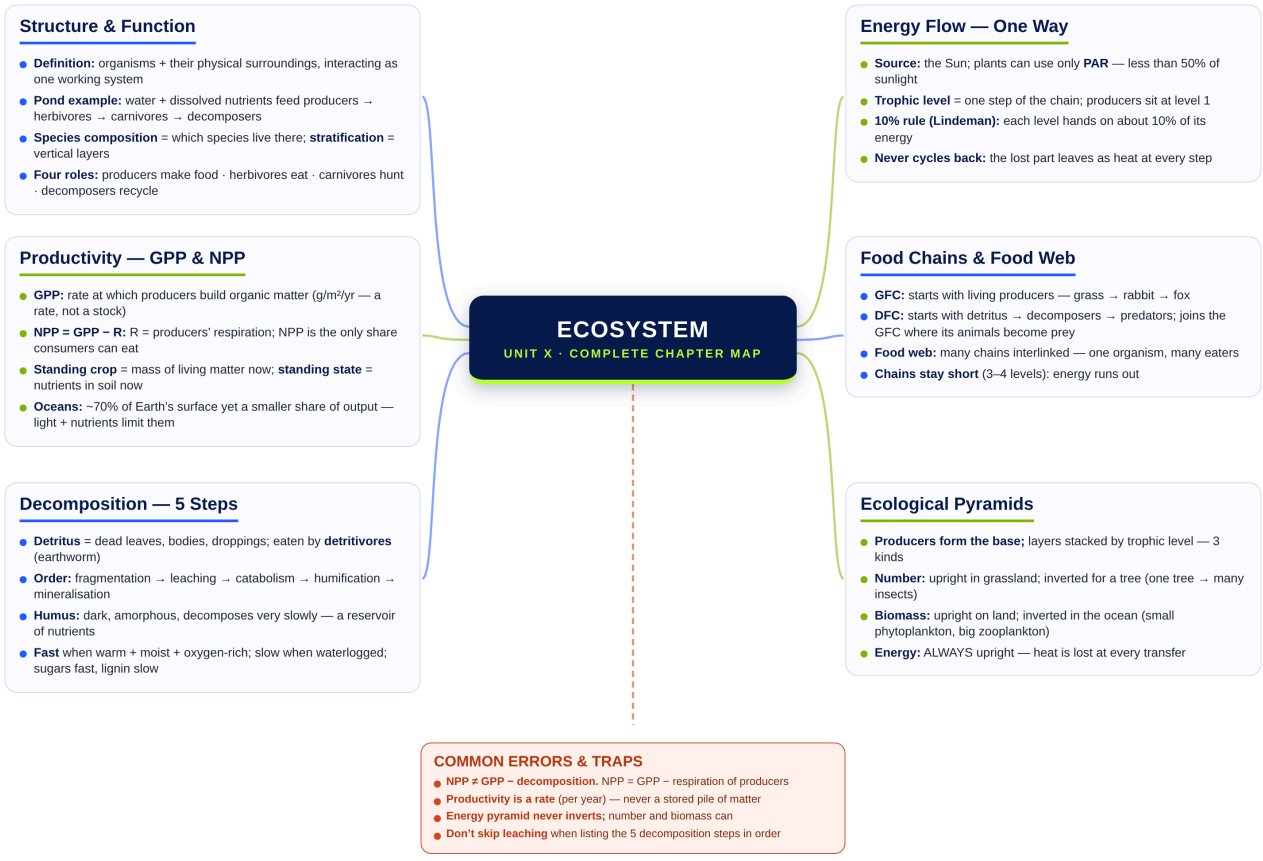


Figure 2: Whole chapter on one page — the six branches of Unit X and the four traps that catch most students.

Practice Questions

Q1. A grassland's producers show a gross primary productivity of 25,000 kJ/m²/yr and respire 9,000 kJ/m²/yr. The net primary productivity is:

(A) 34,000 kJ/m²/yr
(B) 2,500 kJ/m²/yr
(C) 9,000 kJ/m²/yr
(D) 16,000 kJ/m²/yr

Q2. In a grassland, the producers store 40,000 kJ/m²/yr of energy. Following the 10% rule, the energy reaching the fourth trophic level is:

(A) 4,000 kJ/m²/yr
(B) 400 kJ/m²/yr
(C) 40 kJ/m²/yr
(D) 4 kJ/m²/yr

Q3. In decomposition, the step that comes immediately **after** fragmentation is:

(A) Catabolism
(B) Leaching
(C) Humification
(D) Mineralisation

Q4. In a tree ecosystem, the pyramid of number is:

(A) Upright
(B) Inverted
(C) Spindle-shaped
(D) Flat-topped

Q5. The pyramid of biomass is inverted in:

(A) A grassland
(B) A forest
(C) The ocean
(D) A desert

Q6. Photosynthetically active radiation (PAR) makes up about what share of incident sunlight?

(A) About 50%
(B) 1–5%
(C) Nearly 100%
(D) About 25%

Q7. The standing crop of an ecosystem is:

(A) The nutrients held in the soil
(B) The organic matter made per year
(C) The mass of living material at a given time
(D) The dead matter on the forest floor

Q8. Net primary productivity equals:

(A) GPP − respiration
(B) GPP − decomposition
(C) GPP + respiration
(D) GPP ÷ respiration

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Q9. A detritus food chain begins with:

(A) Living green producers **(B)** Herbivores
(C) Top carnivores **(D)** Dead organic matter

Q10. Why do food chains rarely extend beyond 3–4 trophic levels?

(A) Nutrients cycle back to the plants **(B)** Decomposers absorb the leftover energy
(C) Energy is lost as heat at every step **(D)** Producers are too few in number

Solutions

Sol. Q1. The relation is NPP = GPP − R, and respiration is the only subtraction: 25,000 − 9,000 = **16,000 kJ/m²/yr**. Adding the two numbers (34,000) forgets that producers burn food to stay alive.

Answer: (D)

Sol. Q2. Keep only 10% at each step: 40,000 × 10/100 = 4,000 (herbivores); 4,000 × 10/100 = 400 (carnivores); 400 × 10/100 = **40 kJ/m²/yr** at the fourth level.

Answer: (C)

Sol. Q3. The fixed order is fragmentation → leaching → catabolism → humification → mineralisation. Water dissolves the soluble substances right after the pieces are made small — that step is **leaching**.

Answer: (B)

Sol. Q4. One single tree (T1) supports hundreds of fruit-eating insects and birds (T2), so the count grows toward the top — the pyramid of number is **inverted** in a tree ecosystem.

Answer: (B)

Sol. Q5. The standing crop of phytoplankton is small at any instant, yet it multiplies fast enough to feed a larger standing crop of zooplankton — an inverted pyramid of **biomass in the ocean**.

Answer: (C)

Sol. Q6. PAR (photosynthetically active radiation) is the part of sunlight chlorophyll can absorb — less than half of incident sunlight, **about 50%**.

Answer: (A)

Sol. Q7. Standing crop = the **mass of living material** present at a given time, per unit area — a stock, not a rate. The nutrients in the soil are the standing **state** instead.

Answer: (C)

Sol. Q8. NPP = GPP − respiration (R). Decomposition is a separate recycling process — it never enters the productivity subtraction.

Answer: (A)

Sol. Q9. The detritus food chain starts from **dead organic matter** (detritus) — fallen leaves, bodies and droppings — before detritivores and decomposers take over.

Answer: (D)

Sol. Q10. At every transfer about 90% of the energy leaves as heat (second law of thermodynamics), so after 3–4 levels too little remains to feed another population.

Answer: (C)

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